Final Audit Report

Audited Body	
Puro.earth Project Proponent	Fasera Holdings Pty Ltd
Name of Contact for Puro.earth Project Proponent	Tim Munro
Production Facility Operator	Fasera Holdings Pty Ltd
Name of Contact for Production Facility Operator	Tim Munro
Production Facility Name	Fasera Biochar Plant 1
Production Facility ID	607941
Production Facility Location	Kulja, WA Australia

Audit Description (Start)	
Type of Audit	Production Facility Audit and Output Audit
Number of CORCs under Audit	264
Tonnes of dry biochar under Audit	144.77
Reporting Period Covered by Audit	1 March 2022 to 2 November 2023
Objective of Audit Engagement	Provide assurance opinion against requirements of Puro Standard General Rules Version 3.1
Date of Auditor Engagement	2 December 2023
Date of Audit Report Submission	15 August 2024

Audit Outcomes (Finish)	
Production Facility Eligibility	Eligible
Number of CORCs given assurance	268
Tonnes of eligible dry biochar	144.77 (qualified)
CORC conversion factor (CORCs/t dry biochar)	1.857
Reporting Period given assurance	1 March 2022 to 2 November 2023
Calculation Method	Biochar Methodology

Auditing Body	
Auditor	EnergyLink Services Pty Ltd
Lead Auditor	Rodrigo Pardo Patron
Additional Audit Personnel	Thais Monteiro Voll and Brandon Melyadi
Peer Reviewer	Mark Wallace

This document details the nature and scope of the services provided by a member of EnergyLink Services in respect of the eligibility of the CO_2 Removal Supplier Production Facility under the requirements of Biochar Methodology to the Puro Standard General Rules v3.1.

This document is issued to Puro.earth detailing audit procedures conducted and the auditor's opinion in relation to the eligibility of the Production Facility. It should not be used for any other purpose.

Because of the inherent limitations in any internal control structure, it is possible that fraud, error, or non-compliance with laws and rules may occur and not be detected. Further, the audit was not designed to detect all weakness or errors in internal controls so far as they relate to the requirements set out above as the audit has not been performed continuously throughout the period and the procedures performed on the relevant internal controls were on a test basis. Any projection of the evaluation of control procedures to future periods is subject to the risk that the procedures may become inadequate because of changes in conditions, or that the degree of compliance with them may deteriorate.

The audit opinion expressed in this report has been formed on the above basis.

Copies of relevant documentation are available on the Puro.earth website: puro.earth

COMMERCIAL AND CONFIDENTIAL

Version Control Record

Project Number – J0548				
Document File Name	Date Issued	Version	Lead Auditor	Peer Reviewer
20240820 Audit Final Report Fasera vF.0	20 August 2024	vF.0	Rodrigo Pardo Patron	Mark Wallace

Contents

Abbreviation	Description
‘H’	Hydrogen
‘O’	Oxygen
CO_2	Carbon Dioxide
CORC	CO_2 Removal Certificate
C_{org}	Organic Carbon
GHG	Greenhouse Gas
LCA	Life Cycle Assessment
OC	Overcalculation
UC	Undercalculation
The Puro Rules	the Puro Standard General Rules v3.1
The New Puro Rules	the Puro Standard General Rules v4.0 (Edition 2024)
The Biochar Methodology	Edition 2022 v2 Annex A: of the Puro Rules

PART A: Auditor's Report

To: Puro.earth

Dear Sir / Madam,

EnergyLink Services Pty Ltd (EnergyLink Services) was engaged to perform a reasonable assurance audit of Fasera Holdings Pty Ltd's (Fasera) CO_2 Removal calculation from the production of biochar for the period 1 March 2022 to 2 November 2023 against the eligibility requirements of 'the Puro Standard General Rules v3. 1' (hereafter referred to as "the Puro Rules").

Details of Audited Body

Puro.earth Project Proponent	Fasera Holdings Pty Ltd
Production Facility Operator	Fasera Holdings Pty Ltd
Production Facility Name	Fasera Biochar Plant 1
Production Facility ID	607941
Production Facility location	1477 Burakin, Wialki Road, Kulja, WA 6470, Australia

Responsibility of the Audited Body's Management

The management of the audited body is responsible for the application of the requirements of Biochar Methodology of the Puro Standard General Rules Version 3.1' (hereafter referred to as "the Biochar Methodology") in quantifying CO_2 Removal Certificates (CORCs) from the production of biochar, which is reflected in the proof provided to EnergyLink Services.

The management of the audited body is responsible for preparation and presentation of the evidence in accordance with the Biochar Methodology. This responsibility includes the design, implementation, and maintenance of internal controls relevant to the preparation and presentation of proofs that are free from material misstatement, whether due to fraud or error.

Our independence and quality control

EnergyLink Services has complied with the relevant ethical requirements relating to assurance engagements, which include independence and other requirements founded on fundamental principles of integrity, objectivity, professional competence, due care, confidentiality, and professional behaviour. These include all the requirements defined in the *Fortum – Supplier Code of Conduct*[1].

Furthermore, EnergyLink Services maintains a comprehensive system of quality control including documented policies and procedures regarding compliance with ethical requirements, professional standards, and applicable legal and regulatory requirements, in accordance with *ISQC 1 Quality Control for Firms that Perform Audits and Reviews of Financial Reports and Other Financial Information.*

[1] Fortum (2020), Fortum – Supplier Code of Conduct, available at: www.fortum.com/about-us/contact-us/suppliers/code-of-conduct

Our responsibility

EnergyLink Services' responsibility is to express an opinion on the audited body's quantification of CORCs and compliance with the *Puro Rules* based on the procedures we have performed and the evidence we have obtained.

We have conducted a reasonable assurance engagement in accordance with the *Puro Rules* and relevant international standards, as listed below:

- International Standards on Assurance Engagements ISAE 3000 Assurance Engagements other than Audits or Reviews of Historical Financial Information.
- ISQC 1 Quality Control for Firms that Perform Audits and Reviews of Financial Reports and Other Financial Information, and Other Assurance Engagement.

A reasonable assurance engagement in accordance with relevant international standards involves performing procedures to obtain evidence about the Production Facility process controls and quantification of CORCs in accordance with the Puro Rules. The nature, timing and extent of procedures selected depend on the assurance practitioner's judgement, including the assessment of the risks of material misstatement, whether due to fraud or error. In making those risk assessments, we considered internal controls relevant to the audited body's preparation of proofs. We believe that the assurance evidence we have obtained is sufficient and appropriate to provide a basis for our assurance conclusion.

Summary of procedures undertaken

The procedures we conducted in our reasonable assurance engagement included:

- reviewing evidence provided by the audited body;
- assessing the audited body against eligibility criteria;
- conducting interviews and a site visit to validate the evidence provided;
- analysing procedures that the audited body used to gather data;
- testing of calculations that the audited body performed; and
- identifying and testing assumptions supporting the calculations.

Use of our reasonable assurance engagement report

This audit report has been prepared for use by the audited body and Puro.earth for the sole purpose of reporting on the audited body's quantification of CORCs and compliance with the *Puro Rules*. Accordingly, EnergyLink Services expressly disclaims and does not accept any responsibility or liability to any party other than Puro.earth and the audited body for any consequences of reliance on this report for any purpose.

Inherent limitations

There are inherent limitations in performing assurance audits - for example, assurance engagements are based on selective testing of the information being examined - and because of this, it is possible that fraud, error, or non-compliance may occur and not be detected. An assurance engagement is not designed to detect all misstatements, as an assurance engagement is not performed continuously throughout the period that is the subject of the engagement, and the procedures performed are based on a test basis. The conclusion expressed in this report has been formed on the above basis.

Additionally, non-financial data may be subject to more inherent limitations than financial data, given both its nature and the methods used for determining, calculating, and sampling or estimating such data.

Corrective Action Requests / Recommendations

During the audit process, the auditor issued five (5) corrective action requests, seven (7) recommendations, and two (2) suggestions for improvement.

Corrective Action Request 1: Record-keeping for fuel consumption

Fasera purchased fuel in bulk, the delivered volume information was supported by invoices. Fasera tracked internal fuel consumption, such as fuel used in the electricity generator and vehicles, using a diesel meter to measure the total fuel consumed. These records were maintained through manual record-keeping, which were transcribed into a fuel reconciliation spreadsheet. EnergyLink Services requested Fasera to review the fuel input data in their fuel reconciliation spreadsheet. Consequently, Fasera identified double-counting and discrepancies in fuel usage and updated the calculations in the spreadsheet.

Corrective Action Request 2: LCA

Fasera's Life Cycle Assessment (LCA), developed by a third-party using SimaPro software, could not be fully verified by the auditor due to the matters discussed in Recommendation 7. EnergyLink Services requested Fasera to provide its LCA in a format that enabled the auditor to verify inputs, calculations and outputs. Consequently, Fasera provided a new LCA using the Puro.earth LCA template. Additionally, the auditor requested Fasera to review the numbers considered in the updated LCA, for example; C_{org}, bulk density, soil temperature, etc. Fasera consequently reviewed the numbers and supplied a revised LCA.

Corrective Action Request 3: Heat exchanger

Fasera considered the heat generated during the pyrolysis process as a byproduct that was used in a water heat exchanger to produce eucalyptus essential oil. In the LCA calculation, the GHG emissions for the production of biochar were apportioned between the biochar production and the eucalyptus essential oil processes. However, during the site visit, the auditor observed that the heat exchanger was not operational while the biochar production was in progress. Subsequently, Fasera confirmed that both processes (i.e., the biochar and the essential oil production) did not occur concurrently during the audit timeframe. EnergyLink Services requested Fasera to review the calculations. Fasera subsequently allocated all the emissions associated with the pyrolysis process to biochar production.

Corrective Action Request 4: Wooden pallets emissions

In the LCA, Fasera considered the embodied carbon emissions of the pallets. Upon clarification, Fasera confirmed the pallets used are reused. The auditor requested Fasera to review the emissions data related to the wooden pallets in their LCA. Consequently, Fasera excluded the wooden pallet emissions from the LCA.

Corrective Action Request 5: Laboratory Results

Fasera provided four (4) lab results for the reporting period from 1 March 2022 to 2 November 2023. The auditor noted that Fasera used only one (1) of these lab results for the CORCs calculation. Furthermore, the auditor noted the information in the lab result was inconsistent with the rest of the lab tests. This was pointed out to Fasera, who acknowledged the discrepancy and discarded that lab result. The remaining lab results were considered in the updated calculation of CORCs.

Recommendation 1: Measuring Procedures

Finding

The auditor observed that Fasera recorded the biochar production output based on the volume of biochar sold. Fasera maintained invoices detailing the quantities of bulk bags sold, which were either $1m^3$ or $2m^3$ in volume. During the audit, Fasera provided evidence of a 2 m^3 bag weighing 1,063 kg (531.5 kg per m^3), but the auditor observed a 1 m^3 bag weighing 557 kg during the site visit. This represents a diffeerence of 25.5 kg per m^3 (or 4.8%) between the two samples weighed. Fasera considered 540 kg per m^3 for the quantification of biochar produced. Furthermore, Fasera provided laboratory results that varied considerably, including:

- wet bulk density varying between 0.45 g/cm^3 and 0.59 g/cm^3, this represents a difference of 0.14 g/cm^3 (or 31%); and
- dry bulk density varying between 0.21 g/cm^3 and 0.33 g/cm^3, this represents a difference of 0.12 g/cm^3 (or 57%).

These variances[2] had not been considered appropriately by Fasera.

Recommendation

The auditor recommends Fasera implement a procedure to ensure an accurate measurement of the mass (i.e. tonnes) of dry biochar produced is used for its CORCs claim.

Recommendation 2: Moisture Content

Finding

The auditor noted that Fasera used the results from laboratory tests for moisture content in calculating the dry metric tonnes of biochar produced. While it is good practice to use laboratory tests, in this case, the moisture content varied between 44% and 53%[3], and the variability was not considered in the calculation of dry biochar.

Recommendation

The auditor recommends Fasera implement a procedure to determine and appropriately record the moisture content for every batch of biochar that is produced/delivered. These moisture content measurements can be conducted on-site and are expected to be used to calculate the dry biochar mass. Once sufficient records are obtained and the moisture content variability determined, the frequency of moisture testing can be amended to ensure the accurate moisture content is considered for each batch.

Recommendation 3: Diesel meter

Finding

Fasera purchased fuel in bulk to fill the site's fuel tanks and the purchases were appropriately supported by invoices. Fasera tracked internal fuel consumption, such as fuel used in the generator and vehicles, using a diesel meter to measure the total fuel consumed. These records were maintained through manual record-keeping, which were then transcribed into a fuel reconciliation spreadsheet. Whilst there was a reconciliation performed, the auditor found discrepancies in the reconciliation process (refer to Corrective

[2] It is noted that Fasera included a lab result which was excluded from the audit as part of Corrective Action Request 5, which contained a wet bulk density of 0.79 g/cm^3 and a dry bulk density of 0.27 g/cm^3.

[3] It is noted that Fasera included a lab result which was excluded from the audit as part of Corrective Action Request 5, which contained an erroneous moisture content of 53%, where the calculated moisture content from the wet and dry bulk densities was 66%.

Action Request 1). Additionally, since the Fasera facility engages in activities other than biochar production, such as producing eucalyptus essential oil, the auditor found several fuel-using assets that were not involved in the biochar generation and were excluded from the LCA. As such, the auditor considers the measurement of diesel at the nozzle needs to be accurate. The auditor noted that the diesel meter had not undergone calibration since the purchase of the meter.

Recommendation

The auditor recommends that Fasera calibrate the diesel flow meter according to the manufacturer's recommendations and provide evidence for the next audit.

Recommendation 4: Quality Assurance

Finding

The auditor noted that the soil temperature varied amongst the documents provided to the auditor. For example the LCA Report considered 15°C, whilst in the CORC calculations considered 9°C. Fasera informed the auditor that they had used the default temperature from the Puro.earth Biochar Methodology for the final CORCs calculation which was 14.9°C.

Recommendation

The auditor recommends that Fasera enhance its quality assurance procedures to ensure that relevant data and information are input correctly into the calculation formula of CO_2 removal.

Recommendation 5: Laboratory Results

Finding

Because of the findings described in Corrective Action Request **5**, the auditor has issued the following recommendation.

Recommendation

The auditor recommends Fasera utilise the average laboratory results obtained during the reporting period for carbon content, hydrogen, etc. to calculate the number of CORCs.

Recommendation 6: Evidence of No Double Counting

Finding

The auditor noted that Fasera did not have a procedure to include a “no-double counting clause” in its delivery receipts/invoices for biochar sales. Subsequently, Fasera revised its delivery receipt/invoice template to include the no-double counting clause.

Recommendation

The auditor recommends that Fasera enhance its internal sales procedure to ensure that a “no-double counting clause” is included where relevant (e.g. offtake agreement, documentation of the sale or shipment of the product), as per section 5.5.4 in the Biochar Methodology.

Recommendation 7: LCA

Finding A: LCA Software

The auditor observed that Fasera engaged a third-party company to develop its LCA, who used the software; "SimaPro". According to SimaPro's website, the LCA calculation involves accessing various databases, such as EcoInvent v3. However, the auditor was not provided direct access to the SigmaPro software. The third-party consultant provided screenshots to illustrate their workings, but due to copyright restrictions, they did not share the workings performed. This restricted the ability of the auditor to verify the emissions factors and calculations used, including:

- Calculations: the auditor could not verify the accuracy and completeness of calculations used to determine the final CORC figure;
- Emissions Factors: The auditor could not verify the emissions factors used were the relevant emissions factors;
- Input Data: The auditor could not validate that all data was appropriately and accurately included in the LCA calculations; and
- Input Data Units: The auditor identified, during a meeting with the third-party contractor regarding the LCA software, that the input data unit for fuel usage for energy purposes was mistakenly set to *kWh* instead of *litres*. However, it was stressed that the input data units can be easily adjusted within the LCA software and do not directly impact the calculations as the calculations in the software are independent of the units. The auditor notes that all LCA calculations are directly correlated to using the correct units within a calculation. Therefore, using inappropriate units may lead to miscalculations.

Due to the uncertainty regarding the input data units, the emissions factors used, and the calculations used, the auditor could not provide assurance to the accuracy of the inputs used or the calculations in the LCA software. To enable the auditor to verify the input data, data units, emission factors, and calculations used, Fasera provided a new LCA using the Puro.earth LCA template, as mentioned in Corrective Action Request 2.

Finding B: Heat Exchanger

In the LCA calculation, Fasera considered the heat generated during the pyrolysis process as a byproduct used in a water heat exchanger for the production of eucalyptus essential oil. In the LCA calculation, the GHG emissions for the production of biochar were apportioned between the biochar production and the eucalyptus essential oil processes. However, during a site visit, the auditor observed that the heat exchanger was not operational while the biochar production was in progress. Subsequently, Fasera confirmed that both processes (i.e., the biochar and the essential oil production) did not occur concurrently during the audit timeframe. EnergyLink Services requested Fasera to review the calculations and allocate all the emissions associated with the pyrolysis process to the biochar production, as mentioned in Corrective Action Request 3.

Finding C: Wooden pallets emissions

In the LCA calculation, Fasera considered the embodied carbon emissions of the pallets. Upon clarification, Fasera confirmed the pallets used are reused. The auditor requested Fasera to review the emissions data related to the wooden pallets in their LCA. Consequently, Fasera excluded the wooden pallet emissions from the LCA.

Recommendation

The auditor recommends Fasera augment its LCA calculation procedures to ensure that:

- All data, assumptions, and formulae used for the calculation of emissions associated with the biochar life cycle are traceable, transparent, well-documented, complete, and consistent with the supporting evidence, and
- All relevant emissions sources are properly traceable, transparent, well-documented and consistent in the LCA emissions boundary.

This is expected to increase the auditability of the records and reduce the risk of errors.

Suggestion for Improvement 1: Soil Temperature

To enhance the accuracy of the CORCs calculation, the auditor suggests that Fasera use soil temperatures from the biochar application areas instead of relying solely on default soil temperatures. This is particularly relevant as most farms are in Western Australia, which has a Mediterranean-like climate and higher than average soil temperatures compared to the default.

Suggestion for Improvement 2: Biochar yield

To improve the accuracy of $E_{production}$ calculations, the auditor suggests that Fasera use actual figures for biochar production and biomass. These actual figures should then be used to determine the percentage of biochar yield as shown in the Carbon flowchart of the LCA Report, instead of relying on theoretical figures.

Overall Conclusion

Based on the evidence provided, the auditor has provided the following conclusions:

- Production Facility Audit: **Positive conclusion**
- Production Output Audit:
    - **Qualified conclusion** for biochar sold
    - **Adverse conclusion** for CORCs claim

Production Facility Audit

In the lead auditor's opinion, the carbon removal activity performed in the audited CO_2 Removal Supplier's Production Facility met the eligibility requirements of the Puro Standard General Rules Version 3.1.

Production Output Audit

Biochar Sold

In the lead auditor's opinion, subject to the qualification outlined in *Basis for Qualified Conclusion related to Biochar Sold,* and subject to the implementation of Recommendation 1, the quantification of biochar sold is fairly presented, free of material misstatement and has been calculated in accordance with the Puro Standard General Rules Version 3.1.

Basis for Qualified Conclusion related to Biochar Sold

The auditor identified a significant variance in Fasera's records of biochar production output. A summary of the errors identified include:

- The biochar production quantification was based on the volume of biochar sold. Fasera maintained invoices detailing the quantities of bags sold, which were either 1m^3 or 2m^3 in volume. During the audit, Fasera provided evidence that showed:
    - A 2m^3 bag weighted 1,063 kg of biochar (531.5 kg per m^3), and
    - A 1m^3 bag weighted 557 kg of biochar.

  This represents a difference of 25.5kg per m^3 (or 4.8%) between these two bag sizes.
- Laboratory results provided by Fasera showed:
    - wet bulk density varying between 0.45 g/cm^3 and 0.59 g/cm^3, this represents a difference of 0.14 g/cm^3 (or 31%); and
    - dry bulk density varying between 0.21 g/cm^3 and 0.33 g/cm^3, this represents a difference of 0.12 g/cm^3 (or 57%).

These variances had not been considered appropriately by Fasera for the accurate measurement of the mass (i.e. tonnes) of dry biochar produced used for the CORCs calculation.

CORCs claim

In the lead auditor's opinion, due to the matters discussed in *Basis for Adverse Conclusion related to the CORCs claim*, four (4) of the 264 CORCs calculated are not fairly presented, free of material misstatement and have not been calculated in accordance with the Puro Standard General Rules Version 3.1 . Whilst the net error of four (4) CORCs does not represent a material misstatement, the number of corrections made to the CORC claim (via the Corrective Action Requests) is considerable, and the auditor has, in turn, formed an adverse audit opinion.

In view of the above, the lead auditor is able to express a reasonable assurance opinion that, in all material respects, the claim of **268 CO_2 Removal Certificates (CORCs)** by the audited body for the period 1 March 2022 to 2 November 2023 is eligible for creation.

Biochar	CORCs Under Audit	Net Error (CORCs)	Eligible CORCs	Net Error Rate (%)
Total	264	4 UC	268	1.52%

*OC = Overcalculation / UC = Undercalculation

Basis for Adverse Conclusion related to the CORCs claim

During the audit process, the auditor issued five (5) corrective action requests, seven (7) recommendations, and two (2) suggestions for improvement. Additionally, the auditor found significant variances in Fasera's records of biochar production output as indicated in *Basis for Qualified Conclusion related to Biochar Sold.*

Ongoing Issuance and Digital Monitoring, Reporting and Verification

Despite the audit being completed under the Puro Standard General Rules Version 3.1 (Edition 2023), the auditor has considered the requirements of Appendix A of the Puro Standard General Rules v4.0 (Edition 2024) (the New Puro Rules). The auditor considered the Production Facility and the internal processes, controls and systems to form an opinion over the ongoing issuance and digital monitoring, reporting and verification (dMRV).

In the auditor's opinion, the Fasera Biochar Plant 1 Production Facility has not:

- Demonstrated regular industrial operations; and
- Completed a successful performance verification review (i.e. this audit), as the Audit Report yielded a qualified conclusion for biochar sold and an adverse conclusion for CORCs claim.

Consequently, the auditor recommends Fasera to undertake a new Output Audit, under the New Puro Rules prior to be eligible to the ongoing issuance of certificates.

Sincerely,

Rodrigo PARDO PATRON | Director of Engineering
EnergyLink Services Pty Ltd
Lead Auditor
20 August 2024

Part B: Detailed Findings

Audit Findings and Conclusions

Table 1 to Table 4 summarise the findings from the Production Output Audit. As part of the audit procedures, the auditor performed interviews with site representatives and a site visit to the Production Facility. Where possible, the findings from these procedures were used to validate that the eligibility criteria under the methodology had been met, that the proofs and evidence provided by the audited body were accurate, and that the metering used to quantify the Output was appropriate and correctly calibrated.

Eligibility Assessment

Table 1: Eligibility Assessment

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm that the biochar is used in applications other than energy.	Y	The auditor confirmed that the produced biochar subject to this audit was used as soil amendment for agricultural applications.	N/A.
Confirm that the biochar is produced from sustainable forest or waste biomass raw materials.	Y	The auditor confirmed that the biochar was produced from waste biomass from eucalyptus essential oil production based on harvested oil mallees. Even though the sustainable eucalyptus plantations are not certified by FSC, SFI, PEFC, or equivalent, Fasera provided a letter from a professional forester in Western Australia outlining the Fasera's harvest operations and confirming that Mallee biomass used by Fasera comes only from sustainable plantation sources. The auditor validated the credentials of the forester and is satisfied that the biomass is produced from sustainable sources and is a waste product of a production process.	N/A.

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm that the producer demonstrates net-negativity with results from a LCA that shows: – carbon footprint of the biomass production and supply. – emissions from the biochar production process. – carbon footprint of the biochar end use. – cradle to grave.	Finding	Considering the implications outlined in the *Basis for Adverse Conclusion related to the CORCs claim*, and in particular, the outcomes of Corrective Action Request 2, the auditor confirmed that the LCA provided by Fasera included the relevant information on the emissions arising from the different stages of the biochar cradle-to-grave life cycle.	Recommendation 7
Confirm that the biochar production process meets requirements 1.1.4 to 1.1.6 of the Biochar Methodology, namely that: – It has considered the emissions related to the use of fossil fuels (coal, oil, natural gas). – there is no co-firing of fossil fuels and biomass in the same reaction chamber. – the pyrolysis gases are recovered or combusted. – the molar H/C_{org} ratio is less than 0.7.	Y	The auditor verified that whilst the gasification system operated as an auto-thermal process, generating the necessary thermal energy from the processed feedstock, it initially relied on a diesel burner to initiate and stabilise the syngas flame within the reactor. Based on the above, the auditor confirmed that the emissions related to the use of fossil fuels were considered and there is no co-firing of fossil fuels and biomass in the same reaction chamber. The pyrolysis gases are combusted. The average molar H/Corg ratio is 0.41, which is less than 0.7.	N/A
Confirm that measures are taken for safe handling and transport of biochar to prevent fire and dust hazards.	Y	During the site visit, the auditor verified that the biochar, upon exiting the reactor, was transported by a single screw conveyor enclosed within a water-cooling system. This method was employed to lower the temperature of the biochar before being stockpiled and/or packed into bulk bags, available in either $1m^3$ or $2m^3$ volumes. This process avoids adding moisture to the produced biochar.	N/A.

Standing Data

Table 2: Record Keeping

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm that the standing data of the Production Facility and the CO_2 Removal Supplier was collected and checked.	<u>Finding</u>	Except for the errors outlined in the *Basis for Qualified Conclusion related to Biochar Sold*, the auditor confirmed that the standing data of the Production Facility and the CO_2 Removal Supplier was collected and checked.	N/A.

Production Facility Assessment

Table 3: Production Facility assessment

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm the Production Facility Eligibility under the general rules of Puro Standard.	Y	The auditor confirmed that the Production Facility is eligible under the general rules of Puro Standard and that all necessary evidence had been provided.	N/A.
Confirm that the Production Facility demonstrate Environmental and Social Safeguards.	Y	Fasera provided a letter from BJW Agribusiness, an Australian Association of Agricultural Consultants, stating that the activities carried out by the Fasera Group are operating within the regulatory environment relevant to the type of development and the area where it is based. Based on the information outlined above, the auditor confirmed that the CO_2 Removal Supplier showed sufficient evidence to demonstrate that the Production Facility does not cause significant harm to the surrounding natural environment and local communities.	N/A.
Confirm that the Production Facility demonstrate additionality, that the CO_2 removals are a result of carbon finance, and that the project is not required by existing regulations or other obligations.	Y	The auditor confirmed that the CO_2 Removal Supplier showed sufficient evidence to demonstrate that the project meets the requirements of Clause 1.2.3 of the Biochar Methodology.	N/A.

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm that the quantity of biochar produced and sold is documented via appropriate processes.	Finding	The auditor observed that Fasera recorded the biochar production output based on the volume of biochar sold. Fasera maintained invoices detailing the quantities of bulk bags sold, which were either 1m^3 or 2m^3 in volume. During the audit, Fasera provided evidence of a 2 m^3 bag weighing 1,063 kg (531.5 kg per m^3), but the auditor observed a 1 m^3 bag weighing 557 kg during the site visit. This represents a difference of 25.5 kg per m^3 (or 4.8%) between the two samples weighed. Fasera considered 540 kg per m^3 for the quantification of biochar produced. Furthermore, Fasera provided laboratory results per batch of biochar produced that had a wet bulk density varying between 0.45 g/cm^3 and 0.59 g/cm^3 this represents a difference of 0.14 g/cm^3 (or 31%), and a dry bulk density varying between 0.21 g/cm^3 and 0.33 g/cm^3, this represents a difference of 0.12 g/cm^3 (or 57%). These variances had not been considered appropriately by Fasera.	Recommendation 1
	Finding	The auditor noted that Fasera used the results from laboratory tests for moisture content in calculating the dry metric tonnes of biochar produced. While it is good practice to use laboratory tests, in this case, the moisture content varied between 44% and 53%, and the variability was not considered in the calculation of dry biochar.	Recommendation 2
	Finding	The auditor noted that the soil temperature varied amongst the documents provided to the auditor for example the LCA Report considered 15°C, whilst in the CORC calculations considered 9°C. Fasera informed the auditor that they had used the default temperature from the Puro.earth Biochar Methodology for the final CORCs calculation which was 14.9°C.	Recommendation 4
(CONT.) Confirm that the quantity of biochar produced and sold is documented via appropriate processes.	Finding	Fasera provided four (4) lab results for the reporting period from 1 March 2022 to 2 November 2023. The auditor noted that Fasera used only one (1) of these lab results for the CORCs calculation. Furthermore, the auditor noted the information in the lab result was inconsistent with the rest of the lab tests. This was pointed out to Fasera, who acknowledged the discrepancy and discarded that lab result. The remaining lab results were considered in the updated calculation of CORCs.	Corrective Action Request 5 Recommendation 5

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm that metering infrastructure is in place to determine: the production output. the energy use of the Production Facility.	<u>Finding</u>	Fasera purchased fuel in bulk, the delivered volume information was supported by invoices. Fasera tracked internal fuel consumption, such as fuel used in the generator and vehicles, using a diesel meter to measure the total fuel consumed. These records were maintained through manual record-keeping, which were then transcribed into a fuel reconciliation spreadsheet. Whilst there was a reconciliation performed, the auditor found discrepancies in the reconciliation process (refer to Corrective Action Request 1). Additionally, since the Fasera facility engages in activities other than biochar production, such as producing eucalyptus essential oil, the auditor found several fuel-using assets that were not involved in the biochar generation and were excluded from the LCA. As such, the auditor considers the measurement of diesel at the nozzle needs to be accurate. The auditor noted that the diesel meter had not undergone calibration since the purchase of the meter.	Corrective Action Request 1 Recommendation 3

<table>
<tr><th>Requirement</th><th>Requirement Met?</th><th>Verification Remarks</th><th>Corrective Action Request / Recommendations</th></tr>
<tr><td>Confirm the calculations used to quantify emissions from the process. These must account for:
The energy created by the biochar.
The energy source used in the production process.
Cultivating and harvesting of raw materials (forest vs other biomass).
Transporting of raw materials to the Production Facility (based on distance transported and fuel used).</td><td><u>Finding</u></td><td>The auditor observed that Fasera engaged a third-party company to develop its LCA, who used the software; “SimaPro”. According to SimaPro’s website, the LCA calculation involves accessing various databases, such as EcoInvent v3. However, the auditor was not provided direct access to the SigmaPro software. The third-party consultant provided screenshots to illustrate their workings, but due to copyright restrictions, they did not share the workings performed. This restricted the ability of the auditor to verify the emissions factors and calculations used, including:
<u>Calculations:</u> the auditor could not verify the accuracy and completeness of calculations used to determine the final CORC figure;
<u>Emissions Factors:</u> The auditor could not verify the emissions factors used were the relevant emissions factors;
<u>Input Data:</u> The auditor could not validate that all data was appropriately and accurately included in the LCA calculations; and
<u>Input Data Units:</u> The auditor identified, during a meeting with the third-party contractor regarding the LCA software, that the input data unit for fuel usage for energy purposes was mistakenly set to kWh instead of litres. However, it was stressed that the input data units can be easily adjusted within the LCA software and do not directly impact the calculations as the calculations in the software are independent of the units. The auditor notes that all LCA calculations are directly correlated to using the correct units within a calculation. Therefore, using inappropriate units may lead to miscalculations.
Due to the uncertainty regarding the input data units, the emissions factors used, and the calculations used, the auditor could not provide assurance to the accuracy of the inputs used or the calculations in the LCA software. To enable the auditor to verify the input data, data units, emission factors, and calculations used, Fasera provided a new LCA using the Puro.earth LCA template, as mentioned in Corrective Action Request 2.</td><td>Corrective Action Request 2

Recommendation 7</td></tr>
</table>

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
(CONT.) Confirm the calculations used to quantify emissions from the process. These must account for: – The energy created by the biochar. – The energy source used in the production process. – Cultivating and harvesting of raw materials (forest vs other biomass). – Transporting of raw materials to the Production Facility (based on distance transported and fuel used).	<u>Observation</u>	Fasera used theoretical figures for biochar yield to calculate the total biomass utilized and the total pyrolysis gas emissions. To improve the accuracy of $E_{production}$ calculations, the auditor suggests that Fasera use actual figures for biochar production and biomass. These actual figures should then be used to determine the percentage of biochar yield as shown in the Carbon flowchart of the LCA Report, instead of relying on theoretical figures.	Suggestion for Improvement 2
	<u>Finding</u>	Fasera considered the heat generated during the pyrolysis process as a byproduct that was used for a water heat exchanger to produce eucalyptus essential oil. In the LCA calculation, the GHG emissions for the production of biochar were apportioned between the biochar production and the eucalyptus essential oil processes. However, during the site visit, the auditor observed that the heat exchanger was not operational while the biochar production was in progress. Subsequently, Fasera confirmed that both processes (biochar and essential oil production) did not occur concurrently during the audit timeframe. EnergyLink Services requested Fasera to review the calculations. Fasera subsequently allocated all the emissions associated with the pyrolysis process to biochar production.	Corrective Action Request 3
	<u>Finding</u>	In the LCA calculation, Fasera considered the embodied carbon emissions of the pallets, even though they informed us that the pallets are reused. EnergyLink Services requested Fasera to review the emissions data related to the pallets in their LCA. Consequently, Fasera excluded any pallet emissions from LCA.	Corrective Action Request 4 Recommendation 1
	Y	Except when noted above, the auditor confirmed that the calculations used to quantify emissions from the production process accounted for all aspects of the biochar production process. This included the emissions from harvesting the raw materials, transportation of feedstock and packaging materials, the combustion emissions from the production of char, and the energy sources used in the production process. Nevertheless, the auditor has issued a recommendation to enhance the calculation transparency and auditability.	Recommendation 7
Confirm the CO_2 Removal Supplier is able to calculate the CO2 Removal independently.	Y	Except for the errors outlined in the '*Basis for* Adverse *Conclusion related to the CORCs claim*', the auditor reviewed the evidence provided by the audited body and confirmed that the CO_2 Removal Supplier was able to calculate the CO_2 removal independently.	N/A.

Verification of Proofs

Table 4: Verification of proofs and documentation

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Confirm that the standing data for the Production Facility meets the requirements of the Biochar Methodology and is consistent with other evidence.	Y	Except for the errors outlined in the '*Basis for Qualified Conclusion related to Biochar Sold*', the auditor reviewed the data provided by the audited body and confirmed its consistency with both desktop testing and the site visit.	N/A.
Confirm that the necessary proof and evidence documents are maintained by the Production Facility as per Section 5 of the Biochar Methodology[4].	<u>Observation</u>	To enhance the accuracy of the CORCs calculation, the auditor suggests that Fasera use soil temperatures from the biochar application areas instead of relying solely on default soil temperatures. This is particularly relevant as most farms are in Western Australia, which has a Mediterranean-like climate and higher than average soil temperatures compared to the default.	Suggestion for Improvement 1
	<u>Finding</u>	The auditor noted that Fasera didn't have a procedure to include a "no-double counting clause" in their delivery receipts/invoices for biochar sold. Subsequently, Fasera revised its delivery receipt/invoice template to include the no-double counting clause. Considering the above, the auditor noted that Fasera did not have appropriate processes in place to collect and maintain proof as per Section 5 of the Biochar Methodology. As such, the auditor has issued a recommendation to address this for future claims.	Recommendation 6

[4] Information in Section 5 of the Biochar Methodology includes:

- Proof of sustainability of raw material for forest and/or waste biomass.
- LCA data for biomass and biochar production.
- Justification on the soil temperature used for the calculation of the biochar sequestration.
- Proof of product quality, production volume, sales and end use of biochar.
- Proof of no double counting/C positive marketing.

Peer Reviewer Conclusion

Name of the peer reviewer	Mark Wallace
Peer reviewer's credentials	- Bachelor of Systems Engineering (Honours), majoring in Mechanical and Materials – Australian National University. - Certified Performance Measurement and Verification Analyst (PMVA), Efficiency Valuation Organisation (EVO). - Climate Active Registered Consultant. - Certificate IV in Project Management.
Peer reviewer contact details	Email: mark@energylinkservices.com.au Phone: +61 475 894 971
Outcome of the evaluation undertaken by the peer reviewer	Minor amendments to the report.

Appendix A: Table of Site Visit Findings

Table 5: Site visit summary table

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Check that the raw material is of eligible type and sustainably sourced.	Y	The auditor confirmed that the biochar was produced from waste biomass raw materials. Even though the sustainable eucalyptus plantations are not certified by FSC, SFI, PEFC, or equivalent, Fasera was able to provide the necessary evidence to demonstrate that the feedstock used for biochar production was sourced from Mallee biomass processing, was derived from sustainable raw materials.	N/A.
Check that the LCA provided is consistent with observations on site.	Y	Taking into consideration the implications outlined in the '*Basis for Adverse Conclusion related to the CORCs claim,* the auditor could confirm LCA provided was a qualified representation of the Production Facility and used appropriate assumptions where necessary.	Recommendation 7
Confirm that no fossil fuel is used for heating the pyrolysis reactor, and the pyrolysis gases are recovered or combusted.	Y	The auditor confirmed that although most of the thermal energy required by the Production Facility power plant is created from waste biomass, the system relied on diesel to start the process and heat the reactor to the required temperature and pressure. Additionally, diesel is used for electricity generation that powers the facility. Even though fossil fuel is used to create the energy used in the process, when the biochar is co-generated in the single continuous pyrolysis char machine, it becomes a multi-functional unit whereby the biomass is gasified and turned into co-products (i.e., steam and biochar). This meets the eligibility requirements of the Puro Rules.	N/A.
Check that the Production Facility's documentation system is accurate and reliable for recording the quantity of biochar produced and sold.	<u>Finding</u>	Considering the errors outlined in the '*Basis for Qualified Conclusion related to Biochar Sold,*', the auditor confirmed during the site visit that a system was in place to quantify the biochar produced and sold during the reporting period. However, the auditor identified a significant variance in Fasera's records of biochar production output.	N/A.

Requirement	Requirement Met?	Verification Remarks	Corrective Action Request / Recommendations
Check that appropriate metering infrastructure is in place and calibrated correctly to quantify the Production Facility output and the energy use of the Production Facility.	<u>Finding</u>	Fasera purchased fuel in bulk, the delivered volume information was supported by invoices. Fasera tracked internal fuel consumption, such as fuel used in the generator and vehicles, using a diesel meter to measure the total fuel consumed. These records were maintained through manual record-keeping, which were then transcribed into a fuel reconciliation spreadsheet. Whilst there was a reconciliation performed, the auditor found discrepancies in the reconciliation process (refer to Corrective Action Request 1). Additionally, since the Fasera facility engages in activities other than biochar production, such as producing eucalyptus essential oil, the auditor found several fuel-using assets that were not involved in the biochar generation and were excluded from the LCA. As such, the auditor considers the measurement of diesel at the nozzle needs to be accurate. The auditor noted that the diesel meter had not undergone calibration since the purchase of the meter.	Corrective Action Request 1 Recommendation 3
	<u>Finding</u>	The auditor observed that Fasera recorded the biochar production output based on the volume of biochar sold. Fasera maintained invoices detailing the quantities of bulk bags sold, which were either 1m^3 or 2m^3 in volume. During the audit, Fasera provided evidence of a 2 m^3 bag weighing 1,063 kg (531.5 kg per m^3), but the auditor observed a 1 m^3 bag weighing 557 kg during the site visit. This represents a variance of 25.5 kg per m^3 (or 4.8%) between the two samples weighed. Fasera considered 540 kg per m^3 for the quantification of biochar produced. Furthermore, Fasera provided laboratory results per batch of biochar produced that had a wet bulk density varying between 0.45 g/cm^3 and 0.59 g/cm^3 this represents a difference of 0.14 g/cm^3 (or 31%), and a dry bulk density varying between 0.21 g/cm^3 and 0.33 g/cm^3, this represents a difference of 0.12 g/cm^3 (or 57%). These variances had not been considered appropriately by Fasera.	Recommendation 1
Check that appropriate processes are in place to quantify the inputs to the Calculation formula of CO_2 removal for the purpose of Preparing the Output Report and calculating CORCs.	Y	Taking into consideration the implications outlined in the '*Basis for* Adverse *Conclusion related to the CORCs claim'*, the auditor reviewed the evidence provided by the audited body and could confirm that the inputs to the Calculation formula of CO_2 removal had been correctly determined.	N/A.